

OCL expression (Specify an OCL expression for the method that creates a reservation.)

Context TicketCatalog::reserveTrip(client: Client, trip: Trip)

Pre:

1. trip.tickets -> forAll(i:Ticket | i.client.clientID <> client.clientID and i.trip.tripID <> trip.tripID) // Checks that the client does not already have a ticket for the trip
2. (trip.connections -> size() = 1) or ((trip.connections -> size() = 2) and trip.changeWaitTime <= 1080) or ((trip.connections -> size() = 3) and trip.changeWaitTime <= 2160) // checks that the limit for total layover minutes for the trips is respected.

Post:

1. self.tickets -> exists(i:Ticket | i.client.clientID = client.clientID and i.trip.tripID = trip.tripID)
// checks the existence of the newly created ticket inside the ticketcatalog arraylist called tickets
2. self.tickets -> size() = self.tickets@pre -> size() + 1 // checks that the newly created ticket inside the ticketcatalog arraylist called tickets increases its size by 1.
3. trip.tickets -> exists(i:Ticket | i.client.clientID = client.clientID and i.trip.tripID = trip.tripID)
// checks the existence of the newly created ticket inside the trip class arraylist called tickets
4. trip.tickets -> size() = trip.tickets@pre -> size() + 1 // checks that the newly created ticket inside the trip class arraylist called tickets increases its size by 1.

OCL expression (Specify an OCL expression for the class that represents a reservation.)

Context TicketCatalog

Inv :

1. self.tickets -> forAll(i: Ticket | (i.trip.tickets -> select(k: Ticket | k.client.clientID = i.client.clientID) -> size() = 1)) // checks that a ticket to the same trip has not been given to the same client more than once
2. self.tickets -> forAll(i,k: Ticket | i <> k implies i.ticketID <> k.ticketID) // checks that all ids of tickets are unique